CLOUD CHEAT SHEET · AZURE

Azure OpenAI Service

Enterprise-grade access to GPT, embedding, and reasoning models with Azure governance

Overview

Azure OpenAI Service provides REST API access to OpenAI's model family (GPT chat/completion, embeddings, image, and reasoning models) hosted within Azure's compliance boundary, combined with Azure-native identity, networking, and monitoring controls. It is the standard enterprise choice when data residency, RBAC, and private networking are hard requirements.

Deployment Model

	Details
Resource	The Azure OpenAI account (control plane) — one per region/subscription typically
Deployment	A named instance of a specific model version bound to a quota (TPM/RPM)
Model catalog	GPT chat models, embedding models, and specialized reasoning/vision models
Provisioned Throughput	Reserved capacity (PTUs) for predictable latency at scale vs. pay-as-you-go
Content Filters	Configurable moderation layer applied to prompts and completions

Core Capabilities

- **Chat Completions API** — conversational, function/tool-calling, structured JSON output
- **Embeddings API** — vectorizes text for semantic search and RAG pipelines
- **Assistants-style orchestration** — stateful threads with tool use and file retrieval
- **Fine-tuning** — supervised fine-tuning on selected model families for domain adaptation
- **Batch API** — asynchronous, discounted throughput for non-real-time workloads

Security & Governance Controls

- **Private Endpoint / VNet integration** — removes public data-plane exposure
- **Managed Identity + RBAC** — Cognitive Services OpenAI User/Contributor roles instead of API keys
- **Azure Policy** — enforce approved regions, disable public network access at scale
- **Customer-managed keys** — encrypt fine-tuned model weights and stored data at rest
- **Abuse monitoring & content filtering** — default safety layer; modified access requires Microsoft approval
- **Diagnostic logging** — stream to Log Analytics / Sentinel for prompt-level audit trails

RAG & Grounding Pattern

> **Pattern:** Azure AI Search (retrieval) + Azure OpenAI (generation) + system prompt constraining the model to cite only retrieved context. Use "On Your Data" feature for a managed no-code grounding integration, or custom orchestration (LangChain/Semantic Kernel) for fine-grained control.

Cost & Quota Management

	Effect
TPM/RPM quota	Tokens/requests per minute per deployment — request increases via support ticket
Provisioned Throughput Units	Fixed-cost reserved capacity; predictable latency, better for high, steady load
Pay-as-you-go	Token-metered; best for variable/spiky workloads
Batch processing	~50% discount for jobs tolerant of up to 24h turnaround

Common Pitfalls

	Fix
Hardcoding API keys in application code	Use Managed Identity + Entra ID token auth; rotate/revoke keys if ever exposed
No token budget control on chat history	Implement sliding window or summarization to stay under context limits and control cost

Treating content filters as a compliance guarantee	Layer additional business-logic validation; filters reduce but do not eliminate risk
Single-region deployment with no failover	Use multiple regional deployments behind a routing layer for resilience and quota headroom
Not monitoring token usage per application	Tag deployments and use Azure Monitor/cost analysis to attribute spend per workload

Interview / Exam Quick Hits

- Data submitted via API is **not used to train** the base models by default
- Use **system message + tool/function calling** to constrain output format and reduce hallucination
- Content filter severity levels: safe, low, medium, high — configurable per category (hate, violence, sexual, self-harm)
- Regional model availability varies — check quota page before designing multi-region DR
- Prefer **Managed Identity over API keys** for production workloads to avoid key sprawl and rotation overhead